

Supplementary Material for Constraint-Aware Learning over Candidate-Tree Relaxations for Subgraph Cardinality Estimation

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1 IMPLEMENTATION DETAILS

1.1 Non-Tree Path Propagation

For each non-tree query edge, the deterministic statistic propagates reachable candidate sets along the corresponding candidate-tree path. Processing a propagated candidate charges one operation for each entry in its filtered candidate-neighbor list. If the next scan would exceed the per-edge budget B_{cyc} , that non-tree edge is omitted from the aggregate.

The experiments use $B_{\text{cyc}} = 5,000,000$ operations per non-tree edge. Among the 7,380 workload queries containing at least one non-tree edge, all such edges are evaluated for 6,869 queries, and the mean evaluated-edge coverage is 95.6%.

1.2 Graph-Probe Sampling Schedule

At an internal graph-probe state with extension-domain size M and remaining budget b , the number of candidates sampled uniformly without replacement is

$$m = \min \{M, b, \max\{m_{\min}, \lceil \eta M \rceil\}\}. \quad (1)$$

We use $\eta = 0.5$. The minimum sample count is $m_{\min} = 2$ when the filtered candidate space has more than 10,000 candidate vertices or the query has more than 20 vertices; otherwise, $m_{\min} = 4$. The remaining budget is divided as evenly as possible among the unprocessed sampled children. Algorithm 1 gives the weights, termination conditions, and budget accounting.

1.3 Normalization Safeguards and Model Size

For the statistic s_C , the safeguarded logarithm sets $\epsilon = 10^{-12}$ in $\log_e x = \log \max\{x, \epsilon\}$.

For feature j , let $s_j(q)$ be its standard deviation at observed query size q , and let s_j^{all} be its standard deviation over all training rows. Before fitting the log-linear scale trend, each $s_j(q)$ is lower-bounded by $0.05s_j^{\text{all}}$. The fitted scale is then bounded as

$$\sigma_j^{\text{tr}}(q) = \text{clip}(\bar{\sigma}_j^{\text{tr}}(q), 0.25s_j^{\text{all}}, 4s_j^{\text{all}}). \quad (2)$$

If only one query size is present in the training rows, the global training-row mean and standard deviation are used.

The proposed model has 4,677 trainable parameters: 224 in the four grouped branches, 2,276 in the query-size conditioning network, and 2,177 in the shared prediction MLP. This MLP uses a 32-dimensional GELU hidden layer with dropout 0.15.

2 ADDITIONAL DIFFICULT-QUERY RESULTS

Table 1 reports the six query graph sets with the largest q95 values under the proposed estimator. It complements the candidate-tree-gap analysis in the main paper by identifying the query sets that contain most of the remaining large errors.

Table 1: Query graph sets with the largest proposed-estimator q95 values.

Query set	Rows	q50	q95	$> 10^2$	max
YouTube-32	32	1.914	259.962	3	1.93×10^4
WordNet-16	145	4.074	42.582	0	82.95
YouTube-24	66	2.754	39.984	3	3.17×10^5
WordNet-12	220	2.941	33.873	4	176.95
WordNet-20	108	3.379	28.939	1	100.90
patents-32	161	1.949	26.721	5	6.65×10^4

3 TWO-STAGE PROBE PSEUDOCODE

The main paper defines the sampling distributions, probe features, activation condition, and budget semantics. Algorithm 1 presents the complete procedure.

Algorithm 1: Two-stage candidate-space probe

Require: Query Q , candidate space C , candidate tree T , and DP table D_T
Require: Budgets B_T, B_G and tree log count z_T
Ensure: Probe-derived features $(g_N^C, g_I^C, g_M^C, g_M^G, u_G^C, r_M^G)$

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1: procedure CONSTRUCTPROBEFEATURES
2:    $(S_E, S_I, S_F, N_T) \leftarrow (0, 0, 0, 0)$ 
3:   for  $t = 1, \dots, B_T$  do
4:      $\tau \leftarrow \text{TreeValidSample}(Q, C, T, D_T)$ 
5:     if  $\tau$  is complete then
6:        $e \leftarrow \text{ALLEdgesSatisfied}(Q, \tau); i \leftarrow \text{Injective}(\tau)$ 
7:        $f \leftarrow e \wedge i; (S_E, S_I, S_F, N_T) \leftarrow (S_E + e, S_I + i, S_F + f, N_T + 1)$ 
8:     end if
9:   end for
10:   $g_N^C \leftarrow -\log((S_E + 0.5)/(N_T + 1))$ 
11:   $g_I^C \leftarrow -\log((S_I + 0.5)/(N_T + 1))$ 
12:   $g_M^C \leftarrow -\log((S_F + 0.5)/(N_T + 1))$ 
13:   $(g_M^G, u_G^C, r_M^G) \leftarrow (0, 0, 0)$ 
14:  if  $S_F = 0$  then
15:     $(\bar{c}_G, b_G) \leftarrow \text{GRAPHPROBE}(Q, C, B_G)$ 
16:    if  $\bar{c}_G > 0$  then
17:       $g_M^G \leftarrow z_T - \log \bar{c}_G; u_G^C \leftarrow \min\{b_G/B_G, 1\}$ 
18:    else
19:       $g_M^G \leftarrow 0; u_G^C \leftarrow -\min\{b_G/B_G, 1\}$ 
20:    end if
21:     $r_M^G \leftarrow \max(u_G^C, 0) g_M^G / \max(z_T, 1)$ 
22:  end if
23:  return  $(g_N^C, g_I^C, g_M^C, g_M^G, u_G^C, r_M^G)$ 
24: end procedure
25: function GRAPHPROBE( $Q, C, B$ )
26:   $r \leftarrow \arg \min_{u \in V_Q} (|C(u)|, u)$ 
27:  if  $B \leq 0$  or  $|C(r)| = 0$  then
28:    return  $(0, 0)$ 
29:  end if
30:   $K \leftarrow \min(|C(r)|, B); R \leftarrow \text{UNIFORMWOR}(C(r), K)$ 
31:   $(A, b_G) \leftarrow (0, 0)$ 
32:  for  $i = 1, \dots, K$  do
33:     $b_i \leftarrow \lfloor (B - b_G)/(K - i + 1) \rfloor$ 

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34:    $(a_i, c_i) \leftarrow \text{EXTEND}(\{r \mapsto R_i\}, b_i, |C(r)|)$ 
35:    $(A, b_G) \leftarrow (A + a_i, b_G + c_i)$ 
36: end for
37: return  $(A/K, b_G)$ 
38: end function
39: function  $\text{EXTEND}(\phi, b, w)$ 
40:   if  $b \leq 0$  then
41:     return  $(0, 0)$ 
42:   end if
43:   if  $|\phi| = |V_Q|$  then
44:     return  $(w, 1)$ 
45:   end if
46:    $u \leftarrow \text{NEXTVERTEX}(Q, C, \phi); D \leftarrow D_\phi(u); M \leftarrow |D|$ 
47:   if  $M = 0$  then
48:     return  $(0, 1)$ 

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49:   else if  $|\phi| = |V_Q| - 1$  then
50:     return  $(wM, 1)$ 
51:   end if
52:    $m_{\min} \leftarrow 2$  if  $|V_C| > 10,000$  or  $|V_Q| > 20$ ;
53:    $m_{\min} \leftarrow 4$  otherwise
54:    $m \leftarrow \min\{M, b, \max(m_{\min}, \lceil \eta M \rceil)\}; R \leftarrow \text{UNIFORMWOR}(D, m)$ 
55:    $(A, c) \leftarrow (0, 0)$ 
56:   for  $j = 1, \dots, m$  do
57:      $b_j \leftarrow \lfloor (b - c) / (m - j + 1) \rfloor$ 
58:      $(a_j, c_j) \leftarrow \text{EXTEND}(\phi \cup \{u \mapsto R_j\}, b_j, wM)$ 
59:      $(A, c) \leftarrow (A + a_j, c + c_j)$ 
60:   end for
61: return  $(A/m, c)$ 
end function

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